2024 SSMO Accuracy Round Solutions

SMO Team

AR 1: Let a time of day be three-full if exactly three of its digits are 3s when displayed on a 12-hour clock in the *hh*:*mm*:*ss* format. How many seconds of the day are three-full?

Answer: $\boxed{1188}$

Solution: Note that *hh* is a number from 01 to 12, and *mm* and *ss* are both integers from 00 to 59. There are 45, 14, and 1 two-digit integers from 00 to 59 that have 0, 1, and 2 3s, respectively. If *hh* contains a 3, then there are $(45)(1)+(14)(14)+(1)(45)=286$ possibilities. Otherwise, there are $(11)((1)(14)+(14)(1))=308$ possibilities. Since all times on the 12-hour clock appear twice each day, our answer is $2(286+308)=\boxed{1188}$.

AR 2: Equilateral triangle N is inscribed within circle O. A smaller equilateral triangle P is inscribed within N such that the vertices of P lie on the midpoints of the sides of N. The ratio of the area of O to the area of P can be expressed as $\frac{a\pi\sqrt{b}}{c}$, where a, b, and c are positive integers, a and c are relatively prime, and b is not divisible by the square of any prime. Find $a+b+c$.

Answer: $\boxed{28}$

Solution: WLOG, let the radius of O be 1. Then, the side length of N is $\sqrt{3}$, meaning the side length of P is $\frac{\sqrt{3}}{2}$. Thus, the answer is

$$\frac{\pi}{\left(\frac{\sqrt{3}}{2}\right)^2\cdot\left(\frac{\sqrt{3}}{4}\right)}=\frac{\pi}{\frac{3\sqrt{3}}{16}}=\frac{16\sqrt{3}\pi}{9}\implies 16+3+9=\boxed{28}.$$

AR 3: Three distinct integers a, b, and c with $1\le a,b,c\le 10$ are chosen uniformly at random. The probability that $5\mid a!+b!+c!$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

Answer: $\boxed{19}$

Solution: Note that $1!,2!,\ldots,10!$ have residues $1,2,1,4,0,0,\ldots$ when taken modulo 5. We have 2 cases for residues modulo 5: either one is 0, one is 1, and one is 4, or they are all 0. For the first case, we have 2 ways to choose a 1, 1 way to choose a 4, 6 ways to choose a 0, and 6 ways to permute them, giving $2\cdot 6\cdot 6=72$ solutions. For the other case, we have $6\cdot 5\cdot 4=120$ ways to choose the 3 multiples of 5. Thus, the total number of solutions is $120+72=192$, meaning the answer is

$$\frac{192}{10\cdot 9\cdot 8}=\frac{4}{15}\implies 4+15=\boxed{19}.$$

AR 4: Right triangle ABC has a right angle at C and hypotenuse 1. Let points D and E lie on line AC such that $\angle BDC = \angle BEC = 45^\circ$ and points A, D, C, and E lie in that order. Given that $EA = 13DA$, the area of $\triangle ABC$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

Answer: $\boxed{106}$

Solution: Since BDC and BEC are isosceles right triangles, we have $DC = BC = EC$. Since $AD+DC+CE = EA = 13AD$, we have $DC+CE = 12AD \implies DC = CE = 6AD$. So, $AC = AD + DC = 7AD$. From the Pythagorean Theorem, we have

$$AC^2 + BC^2 = AB^2 = 1 \implies (7AD)^2 + (6AD)^2 = 1 \implies AD^2 = \frac{1}{85}.$$

The area of triangle ABC is

$$\frac{AC \cdot BC}{2} = 21AD^2 = \frac{21}{85} \implies 21 + 85 = \boxed{106}.$$

AR 5: Let $ABCD$ be a convex quadrilateral such that $\angle ABC = 120^\circ$ and $\angle ADC = 60^\circ$. If $AB = BC = CD = 5$, the area of $ABCD$ can be expressed as $\frac{a\sqrt{b}}{c}$, where a, b, and c are positive integers, a and c are relatively prime, and b is not divisible by the square of any prime. Find $a+b+c$.

Answer: $\boxed{82}$

Solution: From the Law of Cosines on triangle ABC, we have

$$AC^2 = 5^2 + 5^2 - 2\left(-\frac{1}{2}\right) \cdot 5 \cdot 5 \implies AC = 5\sqrt{3}.$$

Let $AD = x$. From the Law of Cosines on triangle ADC, we have

$$x^2 + 5^2 - 2\left(\frac{1}{2}\right) 5 \cdot x = \left(5\sqrt{3}\right)^2 \implies x^2 - 5x - 50 = 0 \implies x = 10.$$

Finally, from the sine area formula, we have

$$\begin{aligned}[ABCD] &= [ABC] + [ACD] \\ &= \frac{1}{2}\left(5 \cdot 5 \cdot \left(\frac{\sqrt{3}}{2}\right) + 10 \cdot 5 \left(\frac{\sqrt{3}}{2}\right)\right) \\ &= \frac{75\sqrt{3}}{4} \implies 75 + 3 + 4 = \boxed{82}.\end{aligned}$$

AR 6: Three six-sided dice are rolled. Then, the product of the three numbers on the top faces is calculated. The probability that the product is not a perfect square can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

Answer: $\boxed{197}$

Solution: For simplicity, we will count the number of ways of getting a product that is a perfect square. We will do casework on the number of perfect squares rolled. Firstly, there are $2^3 = 8$ ways to roll 3 perfect squares. Next, if two perfect squares are rolled

and the other number is not a perfect square, it is impossible for the final product to be a perfect square. If 1 perfect square is rolled, then the other two numbers rolled must be equal. There are 3 ways to permute the rolls, 2 perfect squares, and 4 non-perfect squares, giving $3 \cdot 2 \cdot 4 = 24$ possibilities. Finally, if no perfect squares are rolled, the numbers must be 2, 3, and 6 in some order, giving another 6 possibilities. In conclusion, the answer is

$$1 - \frac{8+24+6}{6^3} = 1 - \frac{19}{108} = \frac{89}{108} \implies 89 + 108 = \boxed{197}.$$

AR 7: Let a, b, c, and d be positive real numbers such that

$$\begin{aligned} a^2 + d^2 &= b^2 + c^2 = 425, \\ ac + bd &= 375, \text{ and} \\ ab + cd &= 408. \end{aligned}$$

Find the value of $a + b + c + d$.

Answer: $\boxed{56}$

Solution: We have

$$\begin{aligned} (ac+bd)(ab+cd) &= a^2bc + ac^2d + ab^2d + bcd^2 \\ &= bc(a^2+d^2) + ad(b^2+c^2) \\ &= 425bc + 425ad \implies \\ 375 \cdot 408 &= 425(bc+ad) \implies \\ \frac{375 \cdot 408}{425} &= 360 = bc + ad. \end{aligned}$$

Now,

$$\begin{aligned} (a+b+c+d)^2 &= a^2+b^2+c^2+d^2+2(ab+ac+ad+bc+bd+cd) \\ &= (a^2+d^2)+(b^2+c^2)+2(ac+bd)+2(ab+cd)+2(bc+ad) \\ &= 425+425+2\cdot 375+2\cdot 408+2\cdot 360 \\ &= 3136. \end{aligned}$$

Since a, b, c, and d are positive, $a+b+c+d$ is the positive square root, giving $a+b+c+d = \sqrt{3136} = \boxed{56}$.

AR 8: $ABCD$ is a convex cyclic quadrilateral with $AB = 2$, $BC = 5$, $CD = 10$, and $AD = 11$. Let W, X, Y, and Z be the midpoints of sides AB, CD, BC, and DA, respectively. Find $|WX^2 - YZ^2|$.

Answer: $\boxed{21}$

Solution: We will use vectors. For each point P, let $\vec{p}$ denote its position vector. Since W, X, Y, and Z are midpoints, we have

$$\overrightarrow{WX} = \frac{\vec{c}+\vec{d}-\vec{a}-\vec{b}}{2} \quad \text{and} \quad \overrightarrow{ZY} = \frac{\vec{b}+\vec{c}-\vec{a}-\vec{d}}{2}.$$

Since $|\vec{u}|^2 - |\vec{v}|^2 = (\vec{u}+\vec{v}) \cdot (\vec{u}-\vec{v})$, we have

$$WX^2 - YZ^2 = \frac{(2\vec{c}-2\vec{a}) \cdot (2\vec{d}-2\vec{b})}{4} = (\vec{c}-\vec{a}) \cdot (\vec{d}-\vec{b}).$$

Expanding, we have

$$\begin{aligned}2(\vec{c}-\vec{a})\cdot(\vec{d}-\vec{b}) &= 2\left(\vec{a}\cdot\vec{b}+\vec{c}\cdot\vec{d}-\vec{a}\cdot\vec{d}-\vec{b}\cdot\vec{c}\right)\\ &= |\vec{d}-\vec{a}|^2+|\vec{c}-\vec{b}|^2-|\vec{b}-\vec{a}|^2-|\vec{d}-\vec{c}|^2\\ &= AD^2+BC^2-AB^2-CD^2,\end{aligned}$$

since the $|\vec{a}|^2$, $|\vec{b}|^2$, $|\vec{c}|^2$, and $|\vec{d}|^2$ terms cancel. So, the answer is

$$|WX^2-YZ^2| = \frac{\left|AD^2+BC^2-AB^2-CD^2\right|}{2} = \frac{|121+25-4-100|}{2} = \boxed{21}.$$

AR 9: In the game of S-set, there are 3^{12} distinct cards. Each card has 12 traits, each trait is one of three variants A, B, and C, and no two cards have the same variants for all 12 traits. A full house in the game of S-set is defined to be a nonempty hand of cards in which for each trait, all cards in the hand either share the same variant or have all different variants. The number of hands that are full houses can be expressed as $2\cdot 3^a+3^b$, where a and b are positive integers. Find $a+b$.

Answer: $\boxed{34}$

Solution: Note that any two distinct cards differ in some trait, and in that trait all cards in the hand must have different variants. Since there are only 3 variants, a full house contains at most 3 cards. Now, any 2 cards form a full house, since for each trait they either share a variant or have two different variants. In addition, 2 cards uniquely determine a third card that completes a 3-card full house, since in each trait the third variant is forced. So, there are $\binom{3^{12}}{2}$ full house hands of 2 cards. As each 3-card full house is counted by 3 of its pairs, there are $\frac{(3^{12})(3^{12}-1)}{6}$ full house hands of 3 cards. Lastly, note that all one-card hands are full houses. Thus, the answer is

$$\frac{3^{24}-3^{12}}{2}+\frac{3^{24}-3^{12}}{6}+3^{12} = \frac{2}{3}(3^{24})+\frac{1}{3}(3^{12}) = 2\cdot 3^{23}+3^{11} \implies 23+11 = \boxed{34}.$$

AR 10: Bobby is spinning a rigged wheel with three sections labeled A, B, and C. Two integers $0\le x,y\le 50$ are chosen independently and uniformly at random, such that there is an $x\%$ chance the wheel lands on A and a $y\%$ chance the wheel lands on B. Given that the wheel lands on C the first time, the probability that it will land on C if Bobby spins it again can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

Answer: $\boxed{119}$

Solution: For each x,y, the probability of the first spin being C is $\left(1-\frac{x+y}{100}\right)$ and the probability of the first and second spins both landing on C is $\left(1-\frac{x+y}{100}\right)^2$. So, the answer is

$$\frac{\sum_{0\le x,y\le 50}\left(1-\frac{x+y}{100}\right)^2}{\sum_{0\le x,y\le 50}\left(1-\frac{x+y}{100}\right)}.$$

Since $\mathbb{E}\left(\frac{x}{100}\right) = \mathbb{E}\left(\frac{y}{100}\right) = \frac{1}{4}$, we have

$$\sum_{0\le x,y\le 50}\left(1-\frac{x+y}{100}\right) = 51^2\left(1-\mathbb{E}\left(\frac{x}{100}\right)-\mathbb{E}\left(\frac{y}{100}\right)\right) = 51^2\left(1-\frac{1}{4}-\frac{1}{4}\right) = \frac{51^2}{2}.$$

Now,

$$\sum_{0\le x,y\le 50}\left(1-\frac{x+y}{100}\right)^2=\sum_{0\le x,y\le 50}1-2\sum_{0\le x,y\le 50}\left(\frac{x+y}{100}\right)+\sum_{0\le x,y\le 50}\left(\frac{x+y}{100}\right)^2.$$

We have

$$\sum_{0\le x,y\le 50}1=51^2$$

and

$$2\sum_{0\le x,y\le 50}\left(\frac{x+y}{100}\right)=2\cdot 51^2\left(\mathbb{E}\left(\frac{x}{100}\right)+\mathbb{E}\left(\frac{y}{100}\right)\right)=2\cdot 51^2\left(\frac{1}{4}+\frac{1}{4}\right)=51^2.$$

So,

$$\sum_{0\le x,y\le 50}\left(1-\frac{x+y}{100}\right)^2=51^2-51^2+\sum_{0\le x,y\le 50}\left(\frac{x+y}{100}\right)^2=\sum_{0\le x,y\le 50}\left(\frac{x+y}{100}\right)^2.$$

We compute

$$\begin{aligned}\sum_{0\le x,y\le 50}(x+y)^2&=\sum_{0\le x,y\le 50}(x^2+y^2)+2\sum_{0\le x,y\le 50}xy\\&=2\cdot 51^2\left(\mathbb{E}(x^2)\right)+2\left(\sum_{0\le x\le 50}x\right)^2\\&=2\cdot 51^2\left(\frac{\sum_{i=0}^{50}i^2}{51}\right)+2\left(\frac{50\cdot 51}{2}\right)^2\\&=\frac{51\cdot 50\cdot 51\cdot 101}{3}+\frac{50\cdot 51\cdot 50\cdot 51}{2}\\&=50\cdot 51\cdot 51\left(\frac{101}{3}+\frac{50}{2}\right)\\&=\frac{50\cdot 51\cdot 51\cdot 176}{3}.\end{aligned}$$

So, the answer is

$$\frac{\frac{50\cdot 51\cdot 51\cdot 176}{3}}{\frac{100\cdot 100}{\frac{51\cdot 51}{2}}}=\frac{2\cdot 50\cdot 176}{3\cdot 100\cdot 100}=\frac{44}{75}\implies 44+75=\boxed{119}.$$